

Research & Terminology Assignment 2

① Business Analytics

It involves turning data using today's technology into actionable insights. The technology is used for deep investigation to help inform decisions and drive organization performance. You examine historical and current data to find trends, patterns.

It is divided into 4 types.

In my line of work it would assist by collecting data using descriptive analytics where we would have to understand what caused the drop in the service we actually give the customers. Diagnostic analysis will then ask the question why are we losing customers so much particularly to the bakkee brigade. Is it because our parts are expensive, is our labour rate too high, is our service response effective or we are not considerate of the customers business operation. Predictive analytics will then inform us what will happen should we continue this path or if we change the way it's always been done. Prescriptive analytics will answer what can be done next, new strategies that can be implemented that can be of assistance to improve the business. Discounts that can be implemented, 3 hour response times to reduce customer ^{time} ~~downtime~~ of his operation.

Source : <https://www.ibm.com>

2. Data Analytics

Focuses mainly on collection of raw data current and historical, analyze that data and clean, interpret. It is a more technical role as data is analyzed using special technologies using visualization tools to communicate the discoveries. They use programming languages SQL and Python.

In our forklift business data analysis can assist ~~in~~ by ~~assist~~ providing trends and insights about service patterns. By analysing history data, reading and maintenance records, patterns can be identified to remind or alert the user on when the forklift ~~is~~ has a possibility to fail due to overheat, battery failure or require service maintenance. This will help be scheduling service before all these can happen to prevent downtime and avoid engine failures which are expensive towards the maintenance plan. This then increases the forklift life, prevents accidents and LODs caused by using faulty machine.

The collection of data then proves to be a valuable exercise for each company to employ in their business.

Source: <https://www.investopedia.com>

③ Business Intelligence

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This is a tool that is technological that a company uses collect and analyze internal data from spreadsheets and databases. Companies use these to track KPIs, check patterns, analyze customer behavior and make data drive decisions.

Example: We can use this tool to analyze customer rental patterns, like when farmers rent out forklifts, when wholesales are busier to keep the machines ready for when they need to rent so that we are not in shortage and that the units are in good order to avoid down time, this maximising customer satisfaction and increasing our revenue sales for reaching targets.

Source: <https://www.ibm.com>

④ Data-Driven Decision Making

It is about using real information to direct decisions that the business takes. It creates clarity, helps businesses to see problems early and improve in their efficiency.

It uses facts, numbers, studying the data instead of guessing.

Example: When you request a loan at CAPITEC, they don't just give it to you without checking your historical lending and payment history, my income, ~~data~~ defaults, judgments, spending habits ~~to~~ this will determine the likelihood of me paying them back and affordability. Machine learning models assist in this thorough check.

Source: <https://www.simplilearn.com/data-driven-decision-making-article> by Aditya Kumar. 2025

⑤ Big Data volume, velocity, variety

It refers to the massive, connected datasets that old usual data systems cannot handle. The Volume that explains the amount generated, processed and stored. Velocity refers to the speed it is generated and made available and the various types that the organizations must manage.

Example : In Hospitals complex data is used, health records patient data, data from medical aid claims, medical records. These are enormous amount, in different volumes, the speed it needs to be processed also is off urgency to solve analyse drug interactions, it also needs to be highly secured and processed effectively and accurately with speed.

Source : The knowledgeacademy.

6. Descriptive Analysis.

This is an analysis approach that interprets, summaries historical data by answering the what happened question what is currently happening; uses data visualization to identify trends, to transform data into meaningful insights.

Example : When a botique suddenly loses sales, this type of analysis can assist in collecting historical data to check the beginning and check what went wrong in customer sales, competitive behavior and correct from there with proper insights.

Source : Harvard Business Online.

7. Diagnostic Analytics

5.

This is the logical ^{next} step of analysis of determining why did this happen, it tests relationships between two variables, using regression analysis, assisting in gaining insights into the structure of the relationship. This helps to understand why something happened.

Example: Using the previous example of boutique sales drop, checking the demand of boutiques vs new online Temu, SHEIN stores that offer variety, uniqueness and affordability. This then explains customer behaviour, reduced sales.

Source: Harvard Business Online.

8. Predictive Analysis

We use analysis of data to predict future trends and using historical data to determine future potential outcomes, that can help drive strategic decisions.

Example: We can then predict that if we don't change anything about cash flow of this current boutique the future forecasting of the our cashflow for the upcoming year will be like by using the data tools to check the historical data and predicting the future.

Source: Harvard Business Online.

9. Prescriptive Analysis

This type of analysis focuses on the future as it goes beyond explanations, it recommends the course of action, what can be done to solve the problem using machine learning algorithms.

Example : In our boutique example it could be actions that can be done to improve sales, new marketing strategies, merging with other businesses, or better yet sell should things go south, or improve sales should all the strategies put in place succeed.

Source : Harvard business Online.

10. Data literacy refers to how an individual or organization understands, interprets, analyse and use data to achieve their goals. The knowledge of where to look, how to search and what tools can assist them to achieve their goals.

Example : Being able to read a dashboard that shows product delivery performance, ~~etc~~ seeking relevant information, like customer feedback, using those insights to address poor customer service.

Source : www.dataversity.net.

02 Data, Measurement & Reporting

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11 Structured Data

Highly organised information, that follows a strict format. Information that is stored in rows and columns, making it easy to search and readable to the machine.

It is essential for reporting & machine learning because it is clear and in a predictable format.

Examples:

Spreadsheets, Excel and Google sheets.

Relational databases ~~As~~ MySQL, PostgreSQL tables with defined schemas.

Customer databases, Name, address, number & email.

Source : www.ibm.com

12 Unstructured Data

Data that has no strict format, with large datasets.

Contains both text and non-text data both qualitative (social media) and quantitative (figures, dates) data.

Example : Text form data

Email, Text documents, Social media posts

Non-text data form

Image files (JPG, GIF and PNG), Video files & mobile activity.

Source : www.ibm.com.

13. Data Warehouse

8.

They combine, clean and prepare data so that it can be used for data analysis. They store data from databases, business applications and social media.

Example: A retail store sales data can be stored in a data warehouse, information like a date, amount and transaction details in a format. This information will be stored and assigned to the right cells in a relational database table.

Source: IBM.

14. Data Mining

It's a process of analysing large datasets to discover hidden patterns and relationships. It can solve multiple challenges simultaneously and save on budget though assisting businesses in ^{improving} analyses, decision making.

Example: Data mining can be used to provide teams with insights, monitoring data and spotting unusual activities that ~~the~~ can detect fraud, network bridges and product defects. This can also assist in seeing opportunities that have been overlooked in the market by the specific organization.

Source: IBM.

15. Data Visualization

9.

This is a form of represent information graphically. Using visual tools, charts, graphs and maps it present a clear way of seeing trends, patterns and outliers in data and it helps for us to understand.

Example : The World Health Organisation kept us all in the loop by using tracking methods, graphs of the COVID-19, we saw through the news the inclines of the infections, death rates and when people started getting treatment this was all visual through the presentations they did, as they were collecting data from all countries. Visual data makes it easy to read even if you don't understand the real concept of what is going on.

Source : <https://www.tableau.com>

16. Dashboard

It combines high-level measures, real time statuses, and key operational charts into a single view that is visual that will give stakeholders clear visibility into the company's performance.

Example : My manager has an OPS Dashboard report that ^{he} pulls to monitor technicians performances how they are clocking, using their job logging app, and their per-driving patterns. As team leaders we have to use it to communicate it to them and drive efficiency and reduce penalties of harsh driving and possibly accidents as well.

Source : <https://www.tableau.com>

17. Metric

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It is a quantifiable measurement used to track, and assess a certain process, business performance. The measurement in data, metric puts this data into context to track progress overtime against a target.

Example: Our target budget for tyre sales increases by 5% on every quarter. These sales are then measured to see if we are reaching the target or dropping. This is then presented by Parts manager by collecting data for all branches and presented in percentages to show growth or losses.

Source: <https://www.powermetrics.app>

18. Key Performance Indicator (KPI)

It is also a metric that is actionable and gives strategic value. It measures general activity and evaluates success against top-level strategic goals. It is tracked overtime against targets, baselines and deadlines.

Example: There are measurable keyperformance indicators that I am measured on, every 6-9 months we have an evaluation on my performance review checking if all the targets that have been set for me I have met them or not. If not we then review and adjust, If some targets are too high or require assistance those measures are put in place and followed up by the responsible people. For safety meetings as well we use KPIs.

Source: <https://www.investopedia.com>.